

4(a)	<u>image length</u> ; A magnification triangle magnification 250 (nm) ; (8 mm) A 220 (for 7 mm) A 230 (for 7.5 mm) A 270 (for 8.5 mm) A 280 (for 9 mm)	2
4(b)	pathogen ;	1

Question	Answer	Marks
4(c)	ingests / drinks / AW, contaminated water / water containing the pathogen / food washed in contaminated water / contaminated food ; A examples of accidental ingestion e.g. swimming in contaminated rivers A faecal / oral, transmission A bacteria / <i>Vibrio</i> R if described as virus	1
4(d)	A mRNA or messenger RNA <i>any three from:</i> (one strand only needed) to form <u>mRNA</u> / <u>mRNA</u> is single-stranded ; (m)RNA is, used / needed, to produce, <u>subunit</u> (s) / <u>polypeptide</u>(s) ; (only) one strand (of DNA) is the, template / transcribed, strand ; <i>idea that</i> (complementary) copying / transcribing, other DNA strand would not result in, desired / AW, mRNA / polypeptide ;	3
4(e)(i)	<i>any two from:</i> I receptor I active site (mAbs) specific / different, antigen binding sites / binding sites for antigen ; A (each) mAb binds to a, specific / particular, antigen R if antibody described as an enzyme (each mAb has) specific / different, tertiary structure / variable region(s) / primary structure / sequence of amino acids ; binding site and antigen have complementary shapes or ZAC-3 complementary shape to core polysaccharide and lipid A or 2D6 complementary shape to O-polysaccharide ;	2

Question	Answer	Marks
4(e)(ii)	any four from: yes <i>general points</i> (agglutination / motility prevented, so) bacteria less able to, penetrate mucus / attach to intestinal epithelial cells / colonise intestine / AW ; A idea that fewer bacteria able to, attach / colonise intestine less / no, cholera toxin / released ; bacteria passed out in faeces not able to cause disease in others / AW ; ref. to phagocytosis more effective ; e.g. macrophages stimulated to carry out phagocytosis <i>prevention / treatment</i> (to prevent disease) needs to be given, at early stages / before colonising (intestine) ; <u>passive</u> immunity / <u>passive</u> vaccine ; <i>must be in context of treatment or immediate protection</i> <i>idea that</i> in addition to immune response (so increased effect) ; e.g. acts beside, immune system / immune system cells acts before the immune response can become effective ref. to quicker recovery (if a person has cholera) ; useful when, there is antibiotic resistance / antibiotics cannot be given ; <i>specific mAb</i> mAb ZAC-3 may be more effective for cholera caused by, wider range of <i>V. cholerae</i> / AW; mAb ZAC-3 may be useful if exact form of <i>V. cholerae</i> not known ; mAb 2D6, needs to be targeted against specific <i>V. cholerae</i> forms / may not be effective against other <i>V. cholerae</i> forms ; A not all forms of <i>V. cholerae</i> were tested in lab	4

Question	Answer	Marks
4(e)(ii)	AVP ; e.g. (only) short-term protection / not active immunity / not long term ref. difficulty in delivering mAbs to intestine e.g. may be, digested / destroyed, as pass through gut need to find a way to get mAbs to the intestine if given intravenously need to pass through to gut lumen may be able to prevent multiplication of bacteria mAbs have been used successfully for other diseases	

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